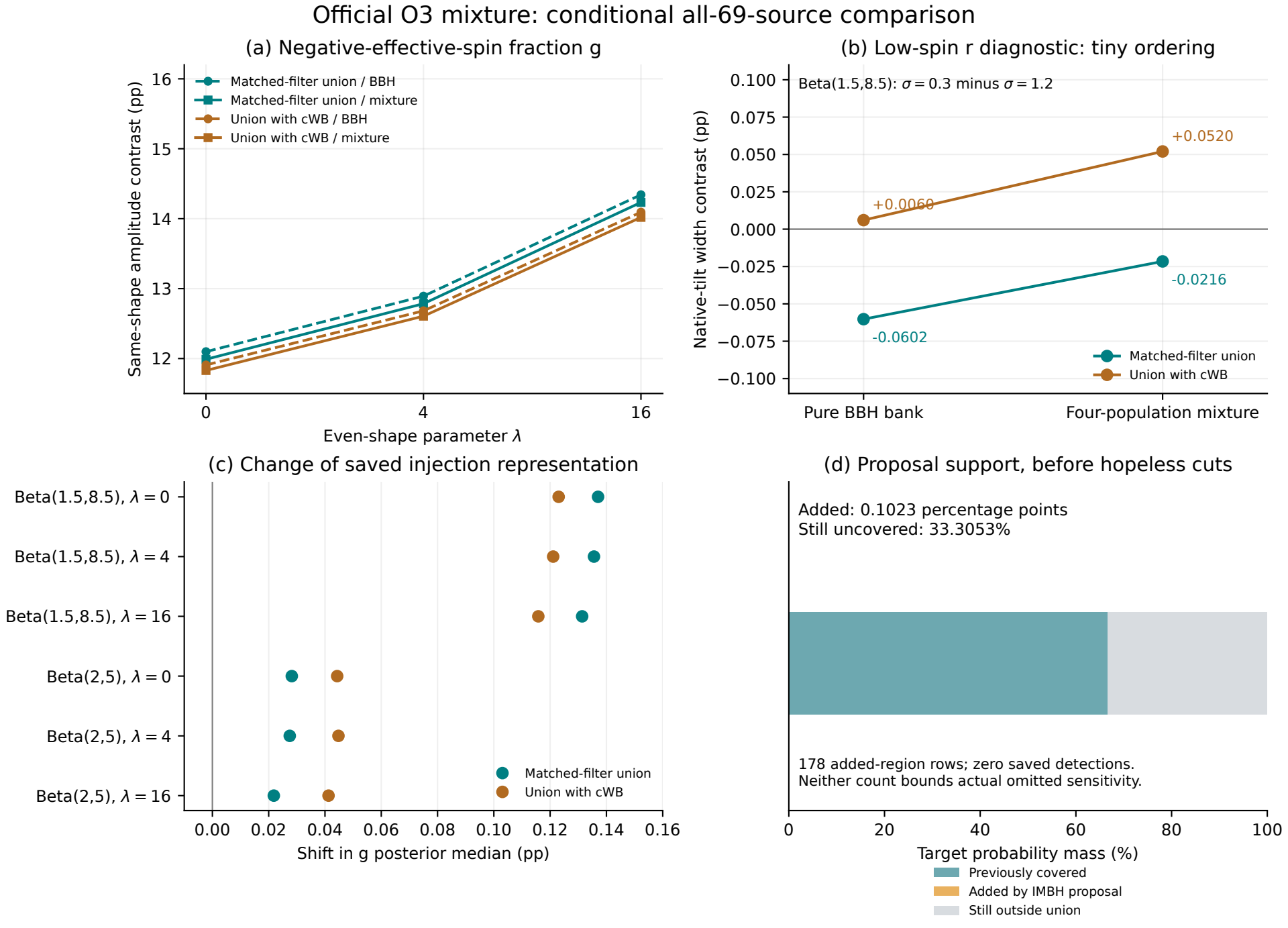

Official O3 mixture: conditional all-69-source comparison
(a) Negative-effective-spin fraction g
Same-shape amplitude contrast (pp)
Even-shape parameter λ
Matched-filter union / BBH
Matched-filter union / mixture
Union with cWB / BBH
Union with cWB / mixture
(b) Low-spin r diagnostic: tiny ordering
Native-tilt width contrast (pp)
Beta(1.5,8.5): σ = 0.3 minus σ = 1.2
+0.0520
+0.0060
-0.0216
-0.0602
Matched-filter union
Union with cWB
Pure BBH bank
Four-population mixture
(c) Change of saved injection representation
Beta(1.5,8.5), λ = 0
Beta(1.5,8.5), λ = 4
Beta(1.5,8.5), λ = 16
Beta(2,5), λ = 0
Beta(2,5), λ = 4
Beta(2,5), λ = 16
Shift in g posterior median (pp)
(d) Proposal support, before hopeless cuts
Added: 0.1023 percentage points
Still uncovered: 33.3053%
178 added-region rows; zero saved detections.
Neither count bounds actual omitted sensitivity.
Target probability mass (%)
Previously covered
Added by IMBH proposal
Still outside union